# From data to transfer decisions

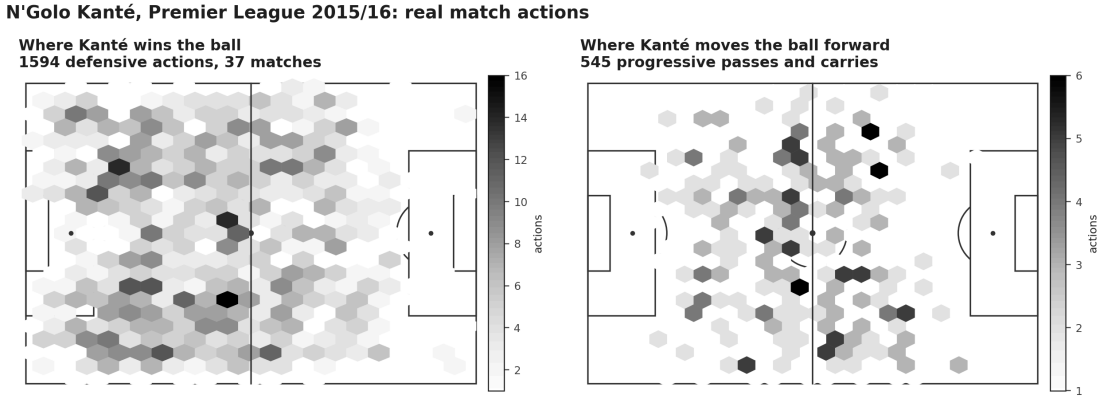
Leicester City, summer 2016. Replace N'Golo Kanté, rebuild the spine, keep the budget.

Dinmukhammed Mynzhassar, AAI-2504M

**380**
real matches parsed

**30,939**
player seasons priced

**3**
signings recommended



# Replace Kanté without losing the identity

## Sign Gueye, Cook and Ighalo for EUR 24m

CM

### Idrissa Gueye

Aston Villa

**EUR 9m**

Fit 95 I Safety 91

CB

### Steve Cook

AFC Bournemouth

**EUR 3m**

Value 100 I Safety 96

ST

### Odion Ighalo

Watford

**EUR 12m**

Fit 79 I Value 82

EUR 50m budget  EUR 26m left for wages and a January buffer  Two year value backtest: +52%

The case for Gueye is tactical fit, not a bargain. The money is made on Cook. Ighalo is the best available box threat.

# What the data can see

Two independent sources, one cutoff date: 1 July 2016.

**380**
Premier League matches

**10,385**
player-match rows

**30,939**
top five player seasons

**1.1m**
transfer records

01
### Parse events
Passes, carries, shots, pressures, defensive actions

02
### Build profiles
Per-90 rates, role clusters, similarity

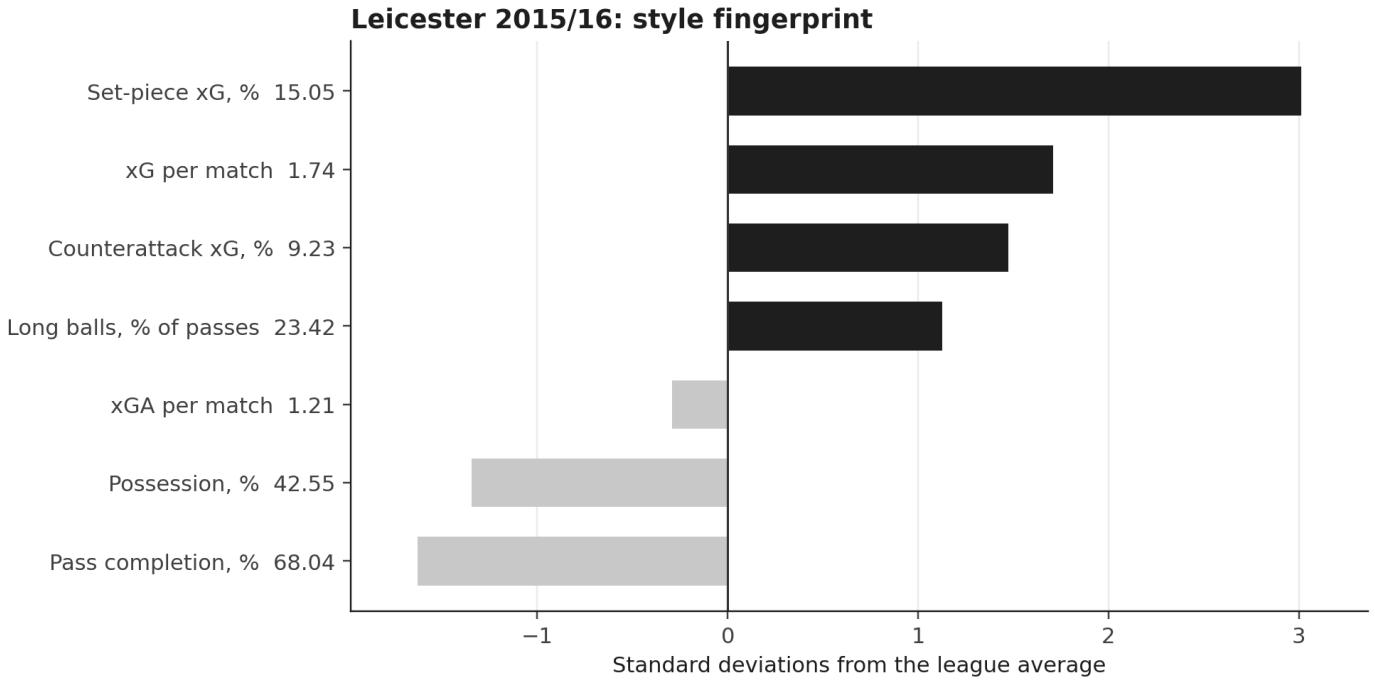
03
### Price the market
Global and local value models

04
### Score fits
Kanté similarity, need profiles, risk

05
### Optimise
Binary plan under the fee budget

Raw data is not committed. The download script and processed tables make the result reproducible.

# What Leicester were




League average is zero. One unit is one standard deviation across 20 teams.

**42.6%**
possession

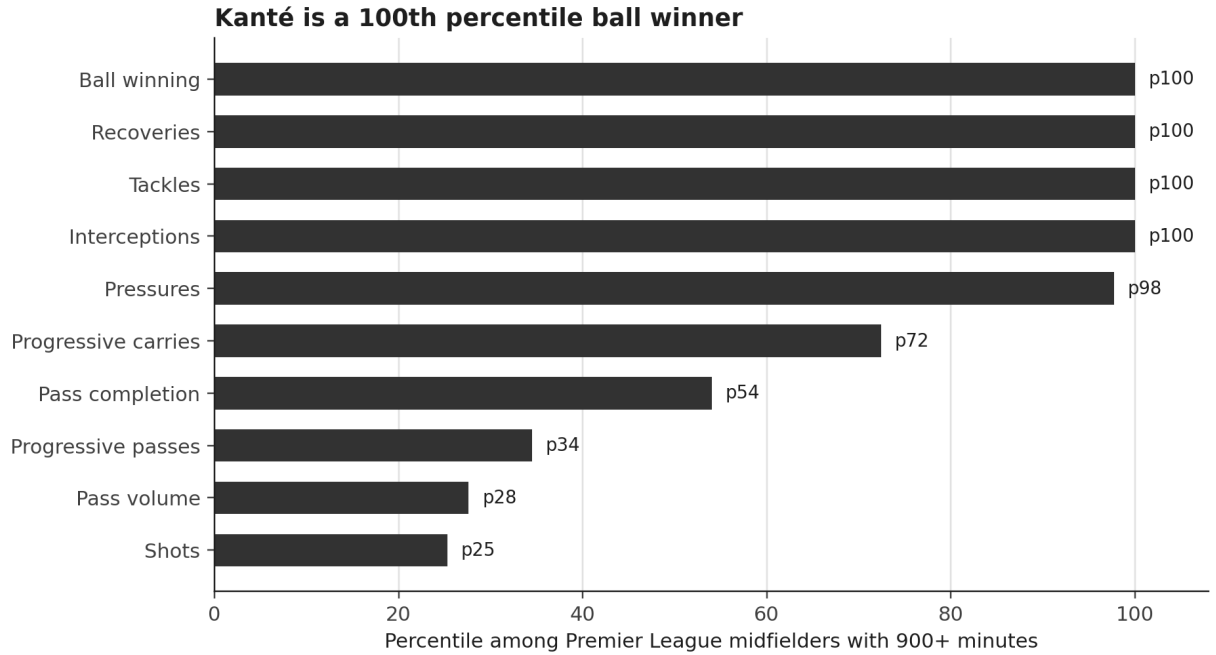
**68.0%**
pass completion

**1.74**
xG per match

**15.1%**
set-piece xG share

Low possession, direct passing, high xG, set pieces and counters. The style did not need a playmaker. It needed a destroyer who could carry the ball.

# Kanté was not a playmaker




Kanté was a 100th percentile ball winner. He was not a high-volume passer.

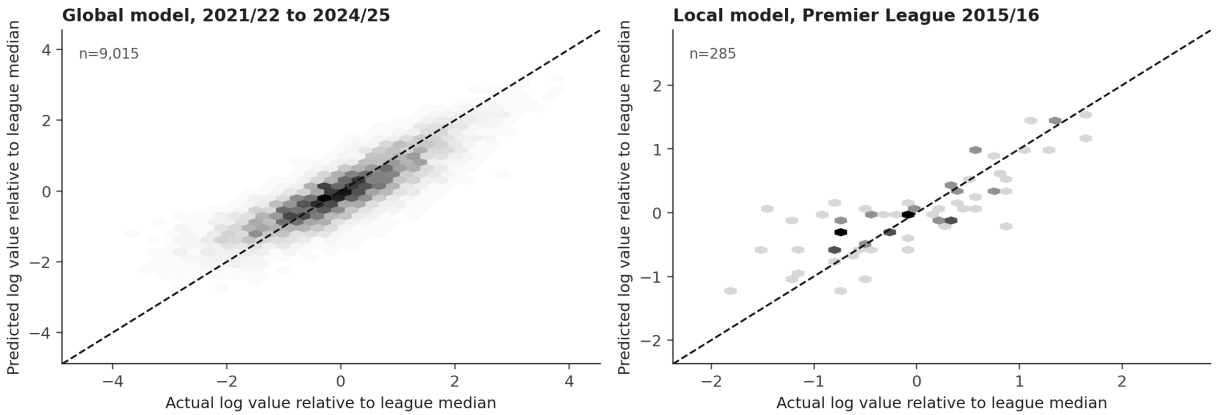
Cosine similarity to Kanté

| | |
|---|---|
| **0.93** | Idrissa Gueye |
| **0.75** | Jan Kirchhoff |
| **0.70** | Victor Wanyama |
| **0.68** | Francis Coquelin |
| 0.00 | Aaron Ramsey, a label warning |

Cosine similarity on ten per-90 actions. Formation labels are not roles.

# Two market models

**Market value models, out of sample**




**0.734**
global R2

**0.855**
global Spearman

**0.614**
local R2

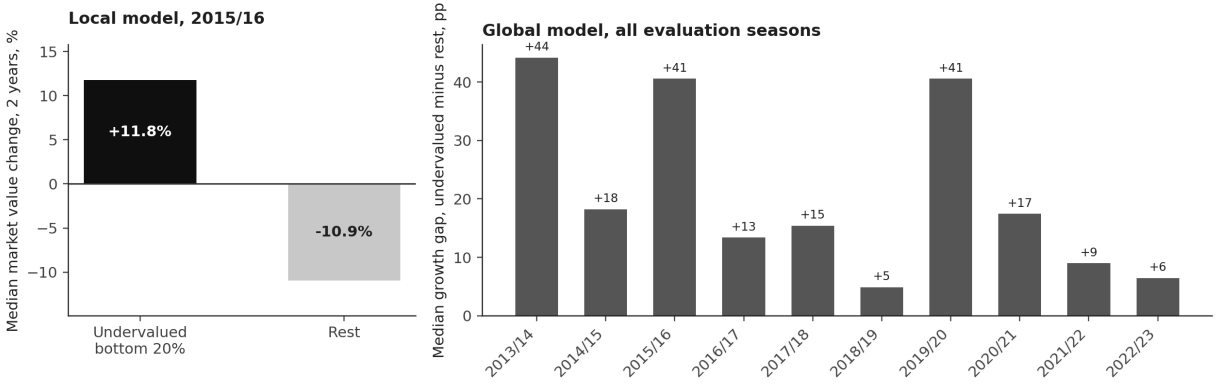
**0.785**
local Spearman

The market pays for goals, assists, passing volume and club reputation. It discounts defensive actions once position and possession are controlled.

Local baseline R2 0.495. The event model adds 0.119. Global test seasons: 2021/22 to 2024/25.

# The undervalued signal is modest but real

**The undervalued signal holds in relative terms**




**+11.8%**
undervalued, 2 years

**-10.9%**
rest, 2 years

**52.8%**
hit rate, undervalued

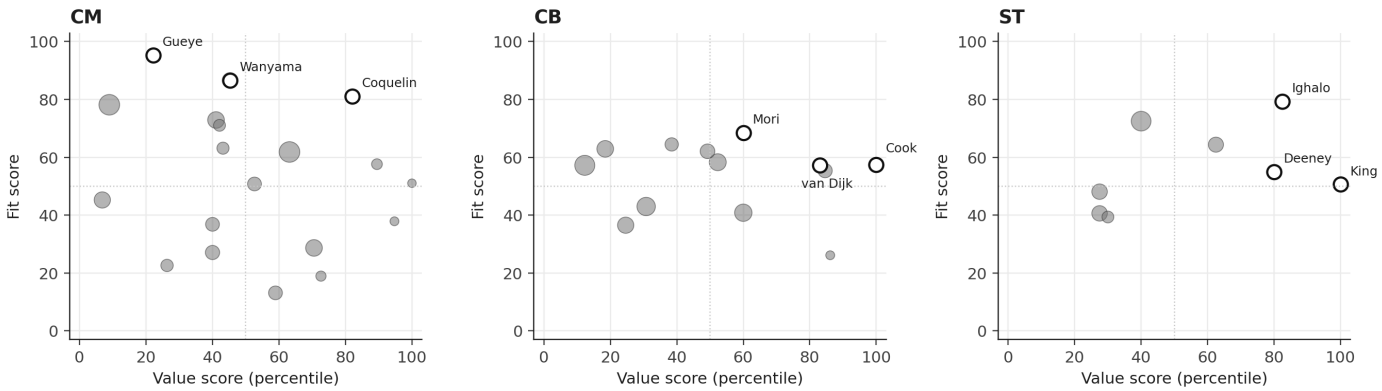
**35.8%**
hit rate, rest

The global model beat the rest in all 10 evaluation seasons on a relative basis. Absolute values fell in several seasons.

This is a signal, not a target. Individual outcomes vary.

# The shortlist

**Candidate map: tactical fit against market value, point size is fee**




| Need | Best balanced | Fee | Fit | Value | Safety | Score |
|---|---|---|---|---|---|---|
| Midfield | Francis Coquelin | EUR 12.0m | 81 | 82 | 74 | 80 |
| Defence | Steve Cook | EUR 3.0m | 57 | 100 | 96 | 76 |
| Attack | Odion Ighalo | EUR 12.0m | 79 | 82 | 96 | 83 |

The shortlist is a trade-off surface, not a single ranking. Cook is the value extreme. Gueye is the fit extreme. Coquelin is the balance.

# The plan

CM

## Idrissa Gueye

Aston Villa

**EUR 9m**

Fit 95 I Value 22 I Safety 91

CB

## Steve Cook

AFC Bournemouth

**EUR 3m**

Fit 57 I Value 100 I Safety 96

ST

## Odion Ighalo

Watford

**EUR 12m**

Fit 79 I Value 82 I Safety 96

Total EUR 24m  Left EUR 26m  Age 25 to 27  All three needs filled

**+80%**
Gueye, 2016 to 2018

**+85%**
Cook, 2016 to 2018

**-23%**
Ighalo, 2016 to 2018

Backtest, not a promise. Two hits and one miss. The plan leaves room for wages, agent fees and a January move.

# Risks and what would change

## What could go wrong

- **Gueye's passing.** A safe short passer, not a creative progressor. Keep Drinkwater as the passer.
- **Cook's step up.** Bournemouth defended deep. Pair him with a progressive partner.
- **Ighalo's outlier.** One strong season at 27. Use him as rotation, not a long contract.
- **Adaptation.** The 4-4-2 pressing triggers need pre-season work.

## What would change the answer

- Budget minus 20%: the recommended plan still fits at EUR 24m.
- Budget minus 40%: the base plan still fits at EUR 27m.
- If the board wants a higher ceiling, Cook can become van Dijk for EUR 33m total.
- If the board wants more passing, Gueye can become Coquelin for EUR 27m.

Limitations: one league of event data, market value as a fee proxy, retrospective outcomes, noisy residuals.

# A decision system, not a black box

## It does not pick a player. It makes the trade-offs visible.

### Reproduce

python scripts/download_data.py
python scripts/build_all.py
python -m streamlit run app.py

### Verify

9 unit tests
Brute-force optimum matches MILP
All report numbers generated from scripts

### Decide

Fit, value and safety are explicit.
The weights are a choice.
The cash buffer is a decision.

AI was used for scaffolding and drafting. The author checked the mapping, the model specification and the recommendation.